Dongryeol Lee

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EDUCATION

Ulsan National Institute of Science and Technology (UNIST) — Mar 2022 – Feb 2028 (expected)

B.S. in Computer Science and Engineering (AI Track); Minor in Business Administration

- **AI & CS coursework:** Natural Language Processing, Machine Learning, Database Systems, Operating Systems, Algorithms, Data Structures, Computer Architecture, Systems Programming, Programming Languages.
- **In progress (Fall 2026):** Deep Learning, Computer Vision, Introduction to Human-Computer Interaction, Introduction to Data Mining.
- **Mathematics:** Applied Linear Algebra, Statistics, Calculus I–II, Differential Equations, Basic Mathematics for AI.
- **Business Administration minor:** Principles of Economics; in progress (Fall 2026): Financial Management, Production & Operations Management.

RESEARCH INTERESTS

- **Vision-Language-Action (VLA) models and embodied AI:** grounded multimodal reasoning, closed-loop decision policies, spatial cognition.
- **World models:** generative dynamics models, physics-grounded representations, simulation, and sim-to-real policy transfer.
- **Foundation models:** vision-language models, diffusion-based trajectory generation, and how long-lived AI systems absorb replaced components such as encoders and world models.

PEER-REVIEWED WORKSHOP PAPERS

1. **SafeCine: Risk-Calibrated Physical Preference Supervision for Language-Conditioned Aerial Cinematography.** Dongryeol Lee (first author), Jaeyeon Bae, and Taehwan Kim (corresponding author). 8th Robot Learning Workshop at NeurIPS 2026. Accepted (poster); non-archival.

RESEARCH EXPERIENCE

Interactive Multimodal Machine Learning Lab (IMML), UNIST — Undergraduate Research Intern (Advisor: Prof. Taehwan Kim) · Dec 2025 – Present

- **SafeCine / RCL-CPM:** Developed Risk-Calibrated Local Counterfactual Preference Mining, which compares candidate trajectories under matched physical perturbations, keeps only pairs resolved by Wilson 95% bounds, and discards ambiguous pairs. Used the resulting preferences for DPO-style denoising alignment of a diffusion proposal policy for language-conditioned camera drones.
- **Isaac Sim (PhysX) benchmark suite:** Built closed-loop physical evaluation environments across 8 scene families (768 unique trajectory rows) in Isaac Sim 4.5. Candidate clearance q0.05 improved by +0.110 m (0.391 m → 0.501 m), tested with parent-stratified bootstrap and exact McNemar tests.
- **Generate–Verify–Select runtime:** Combined proposal-level preference alignment with physical and task admissibility filters and receding-horizon execution.
- **Reported negative result:** The full closed-loop system did not show the expected gain. Documented this with a fail-closed analysis rather than overstating the isolated-test improvement; this proposal–execution gap now motivates follow-up work.

SELECTED PROJECTS

Drive-CoT: Explainable Multimodal Action Generation via Linguistic Mediation — Lead Developer; NLP course research project · May – Jun 2026

- Built a VLA framework that adds autoregressive chain-of-thought reasoning to LLaVA-1.5-7B (LoRA), decoding an explicit spatial rationale before the action tokens.
- Synthesized 10,000 deterministic visual-textual instruction pairs from BDD100K/BDD-X metadata, removing annotation variance while keeping the natural long-tailed class distribution.
- Mitigated majority-class collapse (direct behavior cloning predicted SLOW_DOWN 100% of the time): action accuracy 51.0% → 77.5% (+26.5 pp), minority STOP-class F1 0.75.

Project N.E.O., AI-based Creative Problem Solving (AICP) — Core AI Agent Architecture Engineer; UNIST Undergraduate Research Program · Jun 2026

- Built a multi-agent pipeline with Microsoft AutoGen for ingesting national statistical data (KOSIS), schema validation, and multi-perspective insight synthesis.
- Designed verification protocols between retrieval, code-execution, and summarization agents to enforce factual consistency.

Uncertainty-Guided Monocular Depth Estimation for Embodied Perception — Team project; CSE363 Machine Learning · May – Jun 2026

- Designed monocular depth estimation with epistemic and aleatoric uncertainty modeling to filter noisy depth artifacts under dynamic robotic camera motion.

Variance-Aware Thompson Sampling — CSE30301 Basic Mathematics for AI · Spring 2026

- Bayesian Thompson Sampling with Bessel-corrected variance and a variance floor against early exploration collapse; ranked 1st on all five benchmark environments in the class.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, SQL, Bash, LaTeX
- **Deep learning:** PyTorch, Hugging Face (Transformers, PEFT/LoRA), NumPy, Pandas, scikit-learn
- **Multimodal & agents:** vision-language models (LLaVA-1.5), LoRA fine-tuning, chain-of-thought instruction data, LLM agents (AutoGen)
- **Robotics & simulation:** NVIDIA Isaac Sim 4.5 / PhysX, diffusion trajectory policies, DPO-style preference optimization, closed-loop evaluation
- **Tools:** Linux, Git/GitHub, Docker, Conda, Weights & Biases

LANGUAGES

Korean (native) · English (fluent, academic) · Japanese (basic)

MILITARY SERVICE

Republic of Korea Army · Sergeant · Jul 2023 – Jan 2025 · Completed mandatory service

REFERENCES

Prof. Taehwan Kim — Associate Professor, Department of Computer Science and Engineering, UNIST (IMML Lab advisor)